POLYMERS

Polymer

The word polymer is derived from Greek words, Poly (many) and meros (Parts or unit). A polymer is a large molecule formed by combining small molecules. The individual small molecules from which the polymer is formed are known as "Monomers "and the process by which the monomer molecule are linked to form big polymer molecule is called "polymerization"

Molecular weight of polymer

Due to the varying chain lengths, a polymer is considered to have average molecular mass. Two types of average molecular masses have been recognized namely, number average and weight average molecular masses.

Number average molecular mass: It is obtained by dividing the total weight of the monomers with the number of monomer molecules present in polymer i.e.,

$$M_n = \frac{n_1M_1 + n_2M_2 + n_3M_3}{n_1 + n_2 + n_3}$$

Where n_1, n_2, n_3 are the number of molecules having masses M_1, M_2, M_3 respectively.
The above equation can also be written as, $\overline{M}_n = \frac{\sum niMi}{\sum ni}$

Weight average molecular mass: It gives the representation to different molecular species in proportion to their weight in the given polymer molecule.

$$M_w = \frac{n_1M_1^2 + n_2M_2^2 + n_3M_3^2}{n_1M_1 + n_2M_2 + n_3M_3}$$

The above equation can also be written as, $\overline{M}_w = \frac{\sum niMi^2}{\sum niMi}$

Poly Disparity index [PDI]:

$$PDI = \frac{\overline{Mw}}{\overline{Mn}}$$

If PDI =1 polymer is mono disperse & Homogeneous.
PDI >1 polymer is poly disperse & less Homogeneous.

NUMERICALS

1. In a sample of a polymer, 100 molecules have molecular mass 1000 g/mol, 200 molecules have molecular mass 10000 g/mol, and 200 remaining molecules have molecular mass 100000 g /mol, calculate the number average and weight average molecular mass of the polymer.

Solution:

It is given that,

$n_1 = 100$ & $M_1 = 1000$ g/mol,

$n_2 = 200$ & $M_2 = 10000$ g/mol,

$n_3 = 200$ & $M_3 = 100000$ g/mol.

The number average molecular mass of the polymer is given by

$$M_n = \frac{n_1M_1 + n_2M_2 + n_3M_3}{n_1 + n_2 + n_3}$$

$$= \frac{(100 \times 1000) + (200 \times 1000) + (200 \times 100000)}{100 + 200 + 300}$$

$$= 44200 \text{ g/mol}$$

The weight average molecular mass of the polymer is given by

$$M_w = \frac{n_1M_1^2 + n_2M_2^2 + n_3M_3^2}{n_1M_1 + n_2M_2 + n_3M_3}$$

$$= \frac{(100 \times 1000 \times 1000) + (200 \times 10000 \times 10000) + (200 \times 100000 \times 100000)}{(100 \times 1000) + (200 \times 1000) + (200 \times 100000)}$$

$$= 91407.23 \text{ g/mol}$$

2. In a sample of a polymer, 20% molecules have molecular mass 15000 g/mol, 35% molecules have molecular mass 25000 g/mol, and remaining molecules have molecular mass 20000 g /mol, calculate the number average and weight average molecular mass of the polymer, Calculate PDI and comment on it.

Solution:

It is given that,

$n_1 = 20$ & $M_1 = 15000$ g/mol,

$n_2 = 35$ & $M_2 = 25000$ g/mol,

$n_3 = 45$ & $M_3 = 20000$ g/mol.

The number average molecular mass of the polymer is given by

$$\overline{Mn} = \frac{\sum NiMi}{\sum Ni} = \frac{N_1M_1 + N_2M_2 + N_3M_3}{N1 + N2 + N3 \ldots}$$

$$= \frac{20X15000 + 35X25000 + 45X20000}{20 + 35 + 45} = 20750 \ g/mol$$

The weight average molecular mass of the polymer is given by

$$\overline{Mw} = \frac{\sum Ni\,Mi^2}{\sum Ni\,Mi}$$

$$\overline{M_W} = \frac{N_1M_1^{\ 2} + N_2M_2^{\ 2} + N_3M_3^{\ 2}}{N_1M_1 + N_2M_2 + N_3M_3}$$

$$\overline{M_W} = \frac{20X(15000)^2 + 35X(25000)^2 + 45X(20000)^2}{20X15000 + 35X\,25000 + 45\,X\,20000} = 21385\,g/mol$$

$$\text{Poly dispersity index, PDI} = \frac{\overline{Mw}}{\overline{Mn}} = \frac{21385}{20750} = 1.03$$

PDI >1 ,the given polymer is less homogeneous and poly disperse in nature.

3. A polydisperse sample of polystyrene is prepared by mixing three monodisperse samples in the following proportions. 1g of 10000 molecular weight, 2g of 50000 molecular weight and 2g of 100000 molecular weight. Determine number average and weight average molecular weight. Find the index of polydispersity.

Solution:

It is given that,

$n_1 = 1$ & $M_1 = 10000$ g/mol,

$n_2 = 2$ & $M_2 = 50000$ g/mol,

$n_3 = 2$ & $M_3 = 100000$ g/mol.

The number average molecular mass of the polymer is given by

$$M_n = \frac{n_1M_1 + n_2M_2 + n_3M_3}{n_1 + n_2 + n_3}$$

$$= \frac{(1 \times 10000) + (2 \times 50000) + (2 \times 100000)}{1 + 2 + 3}$$

$$= 62000 \text{ g/mol}$$

The weight average molecular mass of the polymer is given by

$$M_w = \frac{n_1M_1^2 + n_2M_2^2 + n_3M_3^2}{n_1M_1 + n_2M_2 + n_3M_3}$$

$$= \frac{(1 \times 10000 \times 10000) + (2 \times 50000 \times 50000) + (2 \times 100000 \times 100000)}{(1 \times 10000) + (2 \times 50000) + (2 \times 100000)}$$

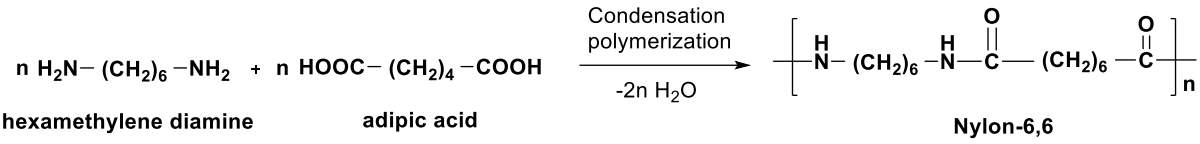
$$= 80967.74 \text{ g/mol}$$

$$\text{PDI} = \frac{Mw}{Mn} = \frac{80967.74}{62000} = 1.30$$

PDI >1, The polymer is polydisperse and less homogeneous

Nylon-6,6

Nylon-6,6 is a polyamide prepared by condensation polymerization of hexamethylenediamine and adipic acid with elimination of water.



Properties:

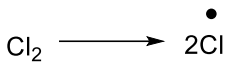
- High melting point (~265 °C).
- High tensile strength and stiffness.
- Excellent toughness and impact resistance.
- High thermal stability.
- Resistant to oils, greases, and many solvents.

3D Printing Applications:

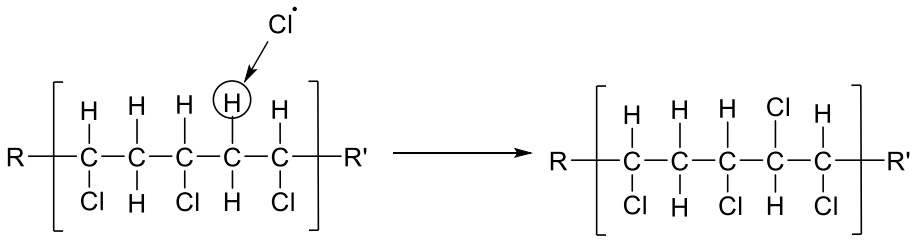
- Nylon-6,6 is used for manufacturing functional prototypes that require high strength and durability.
- It is widely used in the production of gears, bearings, and bushings due to its excellent wear resistance.
- The material is suitable for printing mechanical components such as brackets, fasteners, and housings.
- It is used in automotive parts, including clips, covers, and under-the-hood components.
- Nylon-6,6 is applied in aerospace prototyping for lightweight and high-performance parts.
- It is used for tooling applications, such as jigs, fixtures, and molds.

Chlorinated Poly Vinyl Chloride (CPVC)

Synthesis: CPVC is produced by reacting suspension of PVC resin with chlorine. The reaction is initiated by decomposing chlorine molecule dissolved in water into highly unstable and reactive chlorine atoms.



In the presence of heat or UV the chlorine free radical attacks PVC chain at random location and replaces the hydrogen atom. The liberated hydrogen atom reacts with chlorine to form HCl and chlorine free radical and again the reaction proceeds.



Properties

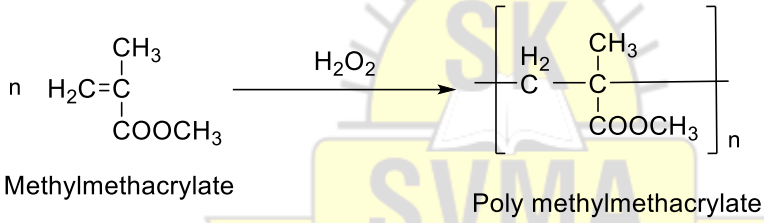
- It has excellent resistance to strong acidic and alkaline conditions.
- It has good thermal resistance than PVC.
- They offer excellent thermal insulation properties.
- They can withstand high temperature compared to PVC.
- They are lightweight and flexible.

Applications

- Employed in switchgear housings, circuit breaker casings, and junction boxes.
- Used for battery casings and enclosures because of chemical resistance.
- Used as insulating material in wires and cables due to excellent dielectric properties.
- Used in heat exchangers and cooling systems.
- Used in fire sprinkler systems due to its flame-retardant nature.

Polymethyl methacrylate (PMMA or Plexiglass)

Synthesis: Polymethyl methacrylate (PMMA) is synthesized by free-radical addition polymerization of methyl methacrylate monomer using initiators such as benzoyl peroxide. The polymerization is carried out by bulk, solution, or suspension methods under heat (about 60–80 °C) in an inert atmosphere.



Properties

- Highly transparent (≈92% light transmission)
- Low density and lightweight
- Good weather and UV resistance.
- Has high tensile strength.
- Resistant to dilute acids and alkalis
- Excellent electrical insulation

Applications

- Lenses and optical fibers.
- Display panels and sign boards.
- Tail lights and indicator cover.
- Transparent roofing sheets.
- Protective covers for instruments.

Conducting polymers

Conducting polymers are a special class of organic polymers that can conduct electricity due to the presence of a conjugated π-electron system along their backbone.

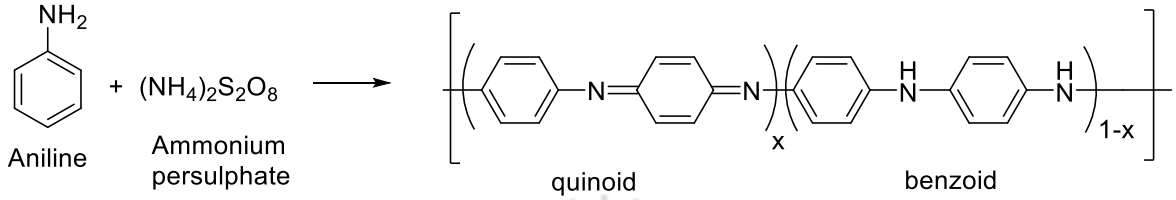
Conducting polymers differ from conventional insulating polymers because of their unique structure containing alternating single and double bonds. This conjugation allows electrons to move freely along the polymer chain, resulting in electrical conductivity. The conductivity of these polymers can be significantly enhanced through a process called doping (oxidation or reduction), which introduces charge carriers such as polarons and bipolarons.

Some well-known examples include Polyaniline, Polypyrrole, and Polythiophene.

Polyaniline

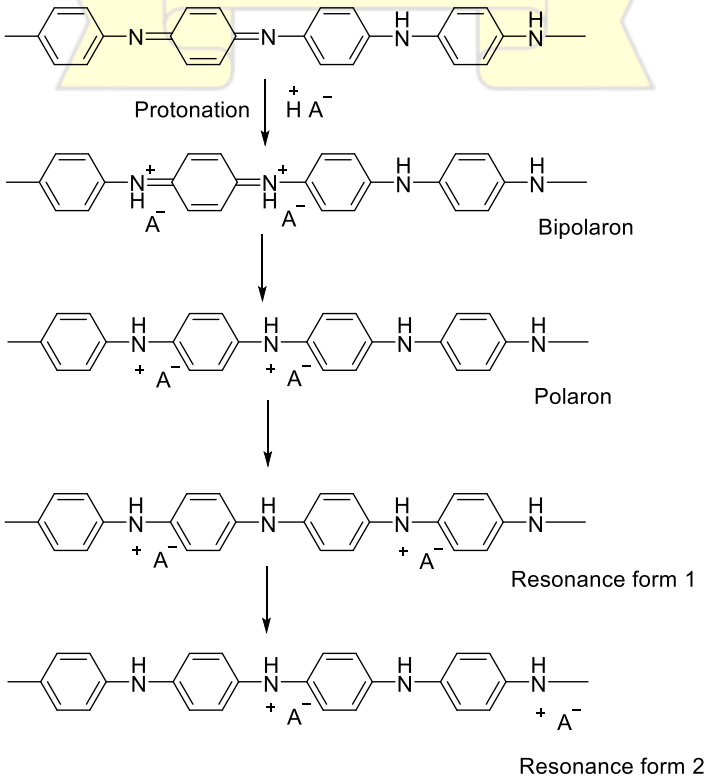
Polyaniline (PANI) is a conducting polymer widely used in electronics and material science.

Synthesis: Polyaniline is synthesized by oxidative polymerization of aniline in the presence of ammonium persulfate. It consists of reduced benzoid units and oxidized quinoid units.



Conduction mechanism of PANI

- PANI has a structure which contains equal proportions of amines and imine sites.
- PANI becomes conductive through protonic acid doping where imine sites are protonated by acids HA to form bipolaron.
- The bipolaron undergoes further rearrangement to form delocalised polaron lattice.
- The delocalized π electrons move along the conjugated chain and hence conductivity arises through π electron delocalization.



Applications in electronic devices

- Used in anti-static coatings - prevents charge buildup on electronic surfaces.
- Used as cathode material in rechargeable batteries.
- Used as hole injection layer in OLEDs.
- Used in corrosion protection paintings as they protect metals through redox activity.
- Used as conductive layer in organic layers solar cells.